

Through the following decade the system was extended throughout the United States. Instrument-flying and navigation by radio beam became standard skills for commercial pilots. By the mid-1930s, passenger aircraft were also almost universally equipped with two-way radios for communication with ground controllers.

Air-traffic control

The control of air traffic around airports became an increasingly urgent concern as traffic levels increased. Arrangements for avoiding collisions were, at first, extremely primitive. Usually, a controller, positioned at a highly visible point in the airport, waved a green or red flag to tell aircraft whether or not it was safe to take off or land. The next refinement involved replacing the flag with a light gun, firing a red or green flare, but these were ineffective in poor visibility.

In 1930 the first air-traffic control tower, equipped with radio, was built at the busy Cleveland Municipal Airport. The pilots of approaching aircraft radioed information on their position to their airline representative at the airport. Controllers used this information to update a map showing where all the aircraft in the vicinity were positioned and radioed pilots if there seemed any risk of a collision. Permission to land or take off was also given by radio. By 1935, there were some 20 airports in the United States operating similar systems.

But traffic kept getting heavier. By the mid-1930s, busy airports such as Newark and Chicago would be handling 60 landings and takeoffs an hour. Aircraft were traveling faster and routinely flying by instruments in poor visibility. Since there was no control over aircraft until they approached an airport, a number of airplanes would arrive at similar altitude in zero visibility, jostling for a chance to land as overburdened controllers strove to avert catastrophe. The situation was not helped by the fact that, at some underfunded airports, the

RIDING THE BEAM

A PRACTICAL PASSENGER air-transportation system required something close to all-weather operation by day and night. Lighted airways were fine to guide aircraft on clear nights, but they were useless in cloud or fog. The answer had to be some form of radio beacon that would remain “visible” to a receiver in the airplane cockpit at all times. But the transmitters available in the 1920s could not generate a beamlike radio signal for an airplane to follow – that would have to wait for the introduction of VHF. The solution was found in the use of the loop antenna, an electrical circuit set upright

from the Earth. It could function as a direction-finding device because a receiver would pick up a strong signal when facing the edge of the loop, but almost none when at right angles to it. Radio engineers experimented with pairs of loop antennae set at right angles to one another, in the shape of a cross. If signals were transmitted alternately from the two antennae, they merged into a single unbroken tone in the equisignal zone – in effect, a radio beam. Pilots could tell whether they were “on the beam” when the humming was steady and continuous. They could also use the “cone of silence” directly above the radio transmitter to fix their position.

It would be an understatement to say that this navigational system was imperfect. The transmissions were sensitive to many forms of interference that could bend or shift the beam, and they were liable to be drowned out by static, especially in the bad weather conditions when they were most needed. But “riding the beam” was still a vast improvement on what had gone before and became standard in the airline business.

GERMAN RADIO CONTROL

Germany was at the forefront of the development of radio navigation devices. This Lufthansa Ju 52 airliner has a direction-finding loop antenna and two-way radio to communicate with the control tower.



NIGHT SERVICE

In the 1930s, Britain's Imperial Airways and France's Air Union (later Air France) operated night passenger services between Le Bourget, Paris, and London's Croydon Airport. Imperial Airways only used British-made aircraft, such as this Handley Page H.P.42. Although slow and ungainly, the H.P.42 afforded a fairly safe and comfortable ride.

KEEPING TRACK

In a United Air Lines flight-dispatch room in the late 1930s, a woman updates information on the progress of flights on a status board. Airline dispatchers kept air-traffic controllers informed of aircraft's whereabouts by telephone.

